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Course: Calculus Beta

Assignment: MyLab Math - Uge 3

Use the arc length formula to find the length of the line segment $y = 5 - 3x$, $0 \leq x \leq 5$.

Notice that the equation of the curve is defined by expressing y as a function of x .

If f' is continuous on $[a, b]$, then the length (arc length) of the curve $y = f(x)$ from the point $A = (a, f(a))$ to the point $B = (b, f(b))$ is as given below.

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Begin by finding the derivative $\frac{dy}{dx}$ of the given function $y = 5 - 3x$.

$$\frac{dy}{dx} = -3$$

The derivative is continuous on $[0, 5]$. Use the above formula to find the length of the given line segment. Find $\left(\frac{dy}{dx}\right)^2$.

$$\frac{dy}{dx} = -3$$

$$\left(\frac{dy}{dx}\right)^2 = 9 \quad \text{Square both sides.}$$

Add 1 to both sides of the equation and simplify.

$$1 + \left(\frac{dy}{dx}\right)^2 = 10$$

Substitute the values of a and b , and the expression for $1 + \left(\frac{dy}{dx}\right)^2$, into the formula.

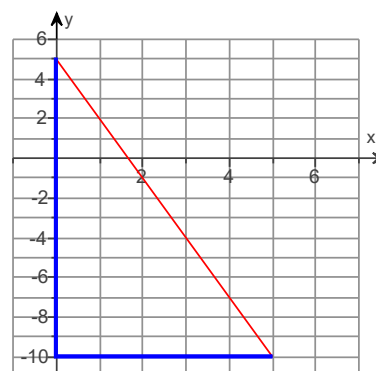
$$\begin{aligned} L &= \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\ &= \int_0^5 \sqrt{10} dx \end{aligned}$$

Now integrate to find the length of the line segment.

$$\int_0^5 \sqrt{10} dx = 5\sqrt{10}$$

Now check the answer by finding the length of the segment as the hypotenuse of a right triangle using the Pythagorean theorem. The graph of the line segment $y = 5 - 3x$, $0 \leq x \leq 5$ is shown in red to the right.

$$\begin{aligned} (\text{hypotenuse})^2 &= (15)^2 + (5)^2 \\ (\text{hypotenuse})^2 &= 250 \\ \text{hypotenuse} &= 5\sqrt{10} \end{aligned}$$



The answer checks. Therefore, the length of the line segment is $5\sqrt{10}$.